

JEE MAIN REVISION MASTER: ATOMS AND NUCLEI

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 47 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Nuclear Physics, Binding Energy & Radioactivity	25 PYQs	Nuclear Physics
Very High	Bohr's Hydrogen Model & Spectral Series	22 PYQs	Atomic Models

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Nuclear Physics, Binding Energy & Radioactivity	$R = R_0 A^{1/3}$, $\Delta E_b = [Z m_p + (A - Z)m_n - M_{\text{nuc}}]c^2$, $N(t) = N_0 e^{-\lambda t}$, $T_{1/2} = \frac{0.693}{\lambda} \approx 0.693\tau$
Bohr's Hydrogen Model & Spectral Series	$r_n = 0.529 \frac{n^2}{Z} \text{ \AA}$, $v_n = 2.18 \times 10^6 \frac{Z}{n} \text{ m/s}$, $E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$, $\frac{1}{\lambda} = R Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Nuclear Physics, Binding Energy & Radioactivity (Very High Yield — 25 PYQs)

Core Principle: Nuclear radius scaling $R \propto A^{1/3}$, constant nuclear density, mass defect conversion (1 amu $\approx$ 931.5 MeV), and radioactive decay law.

■■ **Caution / JEE Pitfall:** Nuclear density is CONSTANT ($\approx 2.3 \times 10^{17} \text{ kg/m}^3$) and completely independent of the mass number A!

■ **Speed Trick:** Amount remaining after n half-lives: $N = (N_0 / 2^n)$, where $n = t / T_{1/2}$.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "Match the following types of nuclei with examples shown $\begin{matrix} \text{Column-I} \\ \text{Column-II} \end{matrix}$ & A. Isotopes &..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Which of the following statements is incorrect with reference of 'Nuclear force'?..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "The ratio of volume of Al^{27} nucleus to its surface area is (Given, $R_0 = 1.2 \times 10^{-15} \text{ m}$)..." [Ans: N/A]

3.2 Bohr's Hydrogen Model & Spectral Series (Very High Yield — 22 PYQs)

Core Principle: Quantization of angular momentum $L = n\hbar$, orbital radii, energy levels of hydrogenic species (Z), and emission lines.

■ **Caution / JEE Pitfall:** Balmer series lines fall in the VISIBLE spectrum ($n_1 = 2$). Lyman series lines fall in the ULTRAVIOLET ($n_1 = 1$).

■ **Speed Trick:** Total number of spectral lines emitted when electron de-excites from state n to ground state is $N = (n(n-1) / 2)$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "An electron transition takes place from excited state to ground state in hydrogen atom, then..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Bohr's second postulate implies quantisation of..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "The radius of first orbit in hydrogen atom is 5.3×10^{-11} m. The kinetic energy E_K , potential energy E_P and total energy..." [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

An electron transition takes place from excited state to ground state in hydrogen atom, then

- (A) Its kinetic energy increases but potential energy and total energy decrease
- (B) Its kinetic energy, potential energy and total energy decrease
- (C) Kinetic energy decreases, potential energy increases but total energy remains same
- (D) Kinetic energy and total energy decrease but potential energy increases

Correct Answer: None

Solution Analysis: For hydrogen atom in the Bohr model, $E_n = -\frac{13.6}{n^2}$ eV where E_n is the total energy of the electron in the n^{th} orbit. When the electron comes from an excited state to the ground state, n decreases. For example, from $n=2$ to $n=1$: $E_2 = -3.4$ eV, $E_1 = -13.6$ eV. So the total energy becomes more negative. Hence, total energy decreases. For an electron in a hydrogen atom, $K = -E$ and $U = 2E$ where K is kinetic energy and U is potential energy. Since total energy E decreases and becomes more negative: Kinetic energy $K = -E$ increases. Potential energy $U = 2E$ decreases and becomes more negative.

Benchmark Problem 2: KCET 2025 (Q#1)

Match the following types of nuclei with examples shown Column-I Column-II
 A. Isotopes i. ${}^7\text{Li}$, ${}^7\text{Be}$ B. Isobars ii. ${}^{18}_8\text{O}$, ${}^{19}_9\text{F}$ C. Isotones iii. ${}^1_1\text{H}$, ${}^2_1\text{H}$

- (A) -ii, B-iii, C-i
- (B) -i, B-iii, C-ii
- (C) -iii, B-ii, C-i
- (D) -iii, B-i, C-ii

Correct Answer: None

Solution Analysis: Here's how the examples match up: Isotopes (same Z , different A) – ${}^1_1\text{H}$, ${}^2_1\text{H}$ → group iii
 Isobars (same A , different Z) – ${}^7_3\text{Li}$, ${}^7_4\text{Be}$ → group i
 Isotones (same N , different Z) – ${}^{18}_8\text{O}$, ${}^{19}_9\text{F}$ → group ii
 So the correct matching is A–iii, B–i, C–ii → Option D.